

minicomputer), you can create a simple workflow with Node-RED. When the temperature rises above a certain level, the cooling system automatically turns on. If the temperature falls too low, the system shuts off. To make the system more reliable, you can set up phone alerts in case the system malfunctions or the temperature goes beyond a safe range. This whole solution can be built with open source tools at a fraction of the cost of proprietary systems, helping you save money on electricity bills and keep your products fresh.

The combination of IoT and open source technology offers endless possibilities, making smart solutions not only affordable but also accessible to people from all walks of life.

Emerging trends in open source IoT

Open source IoT is helping with everything from automating daily tasks to solving complex industrial problems. These developments are not just making life easier but also improving efficiency, saving resources, and supporting smarter decisions in many areas. Let's explore some of the exciting trends shaping the future of open source IoT.

Edge computing: Edge computing is a new way of processing data from IoT devices. Normally, when you use a smart device like a fitness tracker or a security camera, the data it collects is sent to a cloud server far away for processing. This can cause delays, especially when quick responses are needed. For example, if you press a button to turn on your smart lights, and the data has to travel to a distant server and back, the delay can be frustrating.

With edge computing, the device processes data locally, right where it is generated. This means faster responses and fewer delays, which is important for real-time actions. In smart homes, devices like lights, air conditioners, and security cameras respond instantly because they don't need to rely on faraway servers. For example, when you

say, "Turn on the lights," the processing happens on or near the device itself. Today, many households are adopting IoT systems for controlling appliances, managing security, and saving energy, all enhanced by edge computing.

Edge computing is also transforming factories. Machines equipped with this technology can monitor their own performance and detect issues like overheating or wear and tear before a major breakdown happens. This prevents costly repairs and downtime. Companies like Tata Steel and Mahindra & Mahindra are already using these systems to improve efficiency. For instance, Siemens in India uses edge computing in its factories to predict machine maintenance needs, ensuring repairs happen before a complete failure.

This shift is becoming a global trend. A report by Gartner predicts that by 2025, 75% of data generated by businesses will be processed locally at the edge instead of being sent to the cloud. This move not only reduces latency but also enhances data security by keeping sensitive information closer to its source.

Artificial intelligence (AI) and machine learning (ML): When AI and ML are combined with IoT, devices become more than just smart. They can analyse data and make decisions on their own, acting automatically without human input.

AgroStar, a Pune-based company, uses AI and IoT to provide Indian farmers with real-time updates on crop health, guiding them on watering, fertilisation, and harvesting. The sensors monitor factors like soil moisture, weather, and temperature, helping predict crop diseases before they appear. For instance, if dry soil or fungal risks are detected, the system alerts farmers to take preventive action. This helps in boosting yields and reducing crop losses.

In healthcare, wearable devices like fitness trackers and smartwatches are using AI to monitor health in real-time. These devices track metrics like heart rate, blood pressure, and sleep patterns. If

abnormal readings occur, they alert users or even notify doctors. In India, devices like Fitbit and Garmin are popular, and smart platforms in some hospitals use AI and IoT to share patient data with healthcare providers. This ensures faster response times for emergencies like heart attacks or strokes.

AI and IoT combine to provide real-time insights and predictive analysis. AI can process data from IoT devices and recommend actions to improve outcomes.

Blockchain technology for IoT security: With IoT devices becoming a part of daily life—whether it's smart locks, home cameras, or health-monitoring wearables—ensuring the security of data becomes critical. These devices often rely on centralised servers, which can be vulnerable to cyberattacks. Blockchain technology provides an innovative solution by decentralising data storage. Instead of storing data in one place, it is distributed across a network of computers or 'nodes'. This not only strengthens security but also makes it much harder for hackers to manipulate or steal information.

For example, in smart homes, blockchain ensures that only authorised devices, such as a smartphone, can control systems like locks or cameras. If someone tries to hack the system, blockchain detects it and blocks unauthorised access. In India, startups like Oakter and Livpure are incorporating blockchain into their smart products to improve security. A blockchain-enabled smart lock, for instance, can securely verify the identity of authorised family members through a decentralised network.

Blockchain builds trust in IoT by ensuring that devices only interact with verified sources. Its decentralised nature makes it nearly impossible for hackers to alter data or disrupt systems. Industries using blockchain in IoT also benefit from tools like Hyperledger and Ethereum, which are open source and available to developers.

Beyond homes, blockchain is helping secure supply chains in India, particularly in sectors like food and